VaaniQ: Cross-Lingual Transfer, Codec Shift and Calibration in AI-Generated Voice Detection for Hindi, Marathi and Tamil

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***Abstract*—**Audio deepfake detectors are validated largely on English speech, and reported accuracy rarely separates in-domain discrimination from source-independent generalisation. We present VaaniQ, an evaluation framework for AI-generated voice detection in Hindi, Marathi and Tamil. A speaker-disjoint benchmark pairs 900 Kathbath bonafide clips with 900 IndicSynth synthetic clips plus 16 kbps Opus twins, yielding 584 held-out test instances. Two detectors are compared: an acoustic front end (91.61% accuracy, 6.56% EER) and a frozen XLS-R-300M path (92.12%, 6.88%). Four findings: (1) codec effect is model-dependent—acoustic loses 4.46 pp under Opus while XLS-R does not; (2) an English-only control collapses to 54.79% majority-class rate; (3) leave-one-language-out transfer is asymmetric (Hindi 78.83% vs Marathi/Tamil >93%); (4) calibration helped XLS-R (ECE 0.2048→0.0891) but hurt the acoustic path (0.0245→0.0260). Critically, every bonafide clip is Kathbath and every spoof is IndicSynth, so source predicts label perfectly. The 98.48% V2 source probe confirms that source-independent generalisation is not established.

***Index Terms*—**Audio deepfake detection, anti-spoofing, Indian languages, cross-lingual transfer, Opus codec, confidence calibration, self-supervised speech.

I. INTRODUCTION

Neural TTS systems now clone a speaker from seconds of audio, and the resulting clips travel over messaging platforms after codec compression. Any detector for this setting must survive both the synthesis and the channel. Progress has centred on ASVspoof [1], whose bonafide material is English read speech. Architectures from RawNet2 [2] to AASIST [3] report single-digit EERs on that data, but Muller et al. showed order-of-magnitude degradation on found audio [4] and that models can key on silence statistics [5]—classic shortcut learning [6].

Two gaps remain. First, language: India has twenty-two scheduled languages yet benchmarks remain English-centric; IndicSynth [7] addresses this. Second, calibration: high discrimination and poor calibration coexist [8], [9], so accuracy alone is insufficient. Cross-lingual evaluation, codec robustness and calibration are usually studied separately, making it hard to tell if a reported gain reflects genuine detection or corpus recognition.

VaaniQ fills this gap for Hindi, Marathi and Tamil. It is not a new architecture—front ends and heads are conventional—but an integrated evaluation varying language, codec and calibration one at a time. Five questions drive the study: RQ1 (codec), RQ2 (English→Indic), RQ3 (leave-one-language-out), RQ4 (calibration), RQ5 (human, N=0). The key result is negative: both detectors perform well on V1, but V1 pairs one bonafide corpus against one synthetic corpus, making corpus identity a perfect class predictor. We quantify this and restrict every claim accordingly.

II. RELATED WORK

RawNet2 [2] and AASIST [3] established strong baselines on ASVspoof 2019 LA [1], but both inherit its single-corpus English limitation. XLS-R [10] learns cross-lingual representations from 500k hours of multilingual audio; Pascu et al. showed frozen SSL features cut EER from 30.9% to 8.8% across eight deepfake corpora [8]. Tak et al. reported similar benefits from wav2vec 2.0 [11]. We freeze XLS-R to test whether a multilingual representation transfers better under Indic codec shift.

Muller et al.'s in-the-wild evaluation [4] and silence analysis [5] show countermeasures exploiting benchmark-specific cues. Our point is narrower: on a benchmark with one bonafide and one synthetic corpus, a source vs label probe is vacuous because both probes receive identical supervision. Kathbath [12] provides speaker metadata; IndicSynth [7] provides synthetic Indic speech; FLEURS [13] is our external bonafide source. Guo et al. [9] showed temperature scaling repairs overconfidence; Muller et al. [14] found native speakers outperform non-native listeners.

III. CONTRIBUTIONS

Our contributions are methodological: (1) a speaker-disjoint multilingual benchmark for Hindi, Marathi and Tamil with Opus twins; (2) controlled comparison of acoustic vs frozen XLS-R front ends; (3) compression-robustness evaluation on paired clips; (4) English→Indic and leave-one-language-out transfer tests; (5) calibration study showing validation-selected strategies can hurt held-out calibration; (6) analysis of source-label confounding; (7) reproducibility apparatus with frozen manifests.

IV. DATASET AND PROTOCOL

V1 contains 1,800 clips (900 Kathbath bonafide, 900 IndicSynth spoof), balanced at 600 per language, resampled to 16 kHz, capped at 12 s. Adding Opus twins for val/test yields 2,346 instances. Speakers are hashed to train/val/test (70/15/15) ensuring disjointness. Table I shows composition; Table II shows configuration.

TABLE I
Dataset composition.

Partition	Clips	Inst.	Spk.	Bonafide	Role
Train	1,254	1,254	178	54.79%	fitting
Val	254	508	45	—	selection
Test	292	584	44	54.79%	metrics
Total	1,800	2,346	267	—	—

Test: 320 bonafide + 264 spoof. Majority-class = 54.79%.

TABLE II
Experimental configuration.

Parameter	Value
Languages	Hindi, Marathi, Tamil
Sources	Kathbath (900) / IndicSynth (900)
Sampling	16 kHz mono, max 12 s
Compression	16 kbps Opus (ffmpeg libopus)
Acoustic	1024-D embedding + AASIST-compatible head
XLS-R	wav2vec2-xls-r-300m frozen + mean pool
Threshold	0.5 (spoof if $s \geq 0.5$)
Calibration	Temperature scaling, grid search on val

A. Source-Label Confound

V1 has a structural weakness governing the interpretation of every number in this paper: every bonafide is Kathbath and every spoof is IndicSynth, so P(spoof|IndicSynth)=1 and P(bonafide|Kathbath)=1. Any corpus-specific cue—channel, loudness, packaging—is a perfectly valid label predictor on this benchmark. Speaker-disjoint splitting removes speaker leakage across partitions but does nothing about source leakage. Furthermore, the standard source vs label probe is degenerate: with bijective source→label, both probes receive identical supervision and are the same classifier. We report V1 numbers as in-domain held-out performance only.

B. Compression and Metrics

Each val/test clip is encoded with ffmpeg libopus [17] at 16 kbit/s and decoded back to PCM, producing 546 paired twins. Throughout, this is called a WhatsApp-style Opus simulation; it is not audio transported through WhatsApp. All systems use threshold 0.5 with spoof as positive class. We report accuracy, precision, recall, F1, EER (the crossing point where P_fa=P_miss), ROC-AUC, and min-DCF [15] with P_tar=0.05. With this setting, false alarms carry 19× the weight of misses, explaining why low EER and poor min-DCF can coexist. Calibration uses ECE with 15 bins.

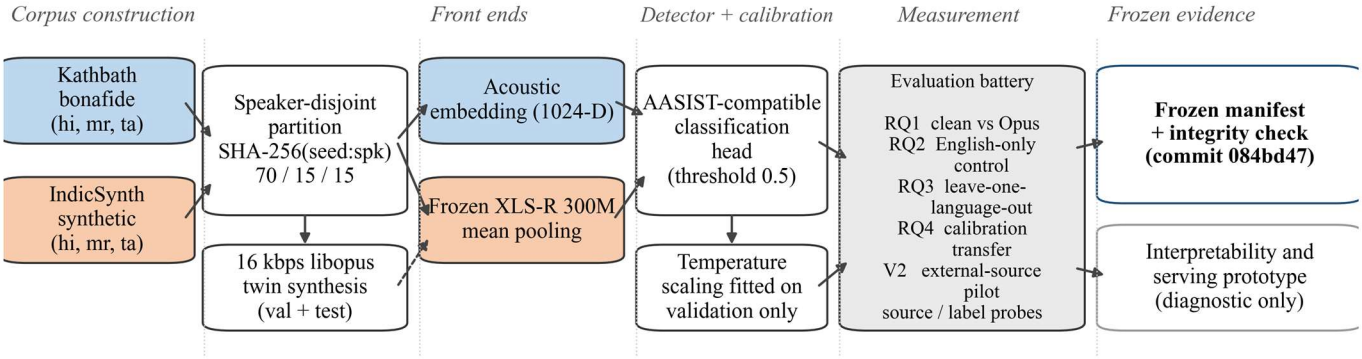


Fig. 1. VaaniQ architecture: two front ends share a classification head. Calibration fits on validation only.

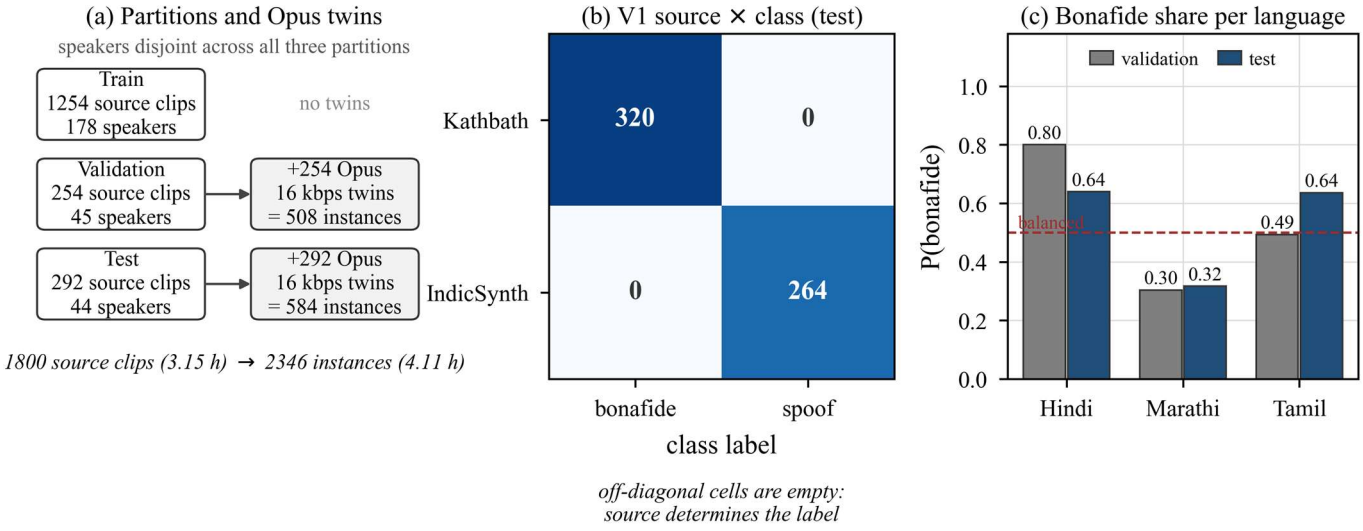


Fig. 2. (a) Partitions and Opus expansion. (b) Source×class: off-diagonal cells are empty, so corpus determines label. (c) Per-language bonafide share differs between val and test.

V. DETECTION MODELS

The acoustic path computes a 1024-D embedding and feeds it to an AASIST-compatible head. The head is not canonical AASIST [3] and should not be compared with published numbers. Checkpoint selection used validation accuracy; test was excluded.

The XLS-R path freezes facebook/wav2vec2-xls-r-300m, mean-pools the last-layer states, and trains only the head. XLS-R was pretrained on 500k hours of multilingual speech [10]. We test whether it behaves differently under Indic codec shift. Head and threshold match the acoustic path.

Three baselines complete the comparison: LFCC-GMM [16], a RawNet2-style approximation (not faithful RawNet2), and an English-only control trained on ASVspoof 2019 LA.

VI. CALIBRATION

Post-hoc calibration divides the logit by temperature T before softmax. Because the transform is monotone, accuracy/EER/AUC are invariant; only ECE and Brier change. Five strategies are compared: uncalibrated, global T, per-language, per-condition, per-language-and-condition. The strategy with lowest validation ECE is shipped; we report test ECE pre- and post-calibration.

VII. RESULTS

A. Overall Comparison

Table III reports all five systems on the 584-instance test set. Three systems (LFCC-GMM, RawNet2-style, English-only) score exactly 54.79%—the majority-class rate. Each predicts bonafide for every instance, so accuracy is uninformative. Only their EER and AUC columns matter: LFCC-GMM ranks above chance, RawNet2-style is near chance, English-only is anti-correlated.

TABLE III
Primary comparison (n=584). Brackets: 95% bootstrap CI.

System	Acc	Prec	Rec	F1	EER	AUC	mDCF
Acoustic	91.61	85.48	98.11	91.36	6.56 [4.6,9.2]	0.9729	0.784
XLS-R	92.12	88.38	95.08	91.61	6.88 [4.8,9.4]	0.9828	0.314
LFCC-GMM	54.79	0	0	0	23.5 [18.8,28.2]	0.820	0.50
RawNet2-approx	54.79	0	0	0	43.2 [39.2,47.2]	0.585	0.96
English-only	54.79	0	0	0	76.6 [73.2,80.1]	0.162	1.06

Three degenerate systems predict bonafide for all instances.

The two main detectors are close but inconsistently ordered. XLS-R leads on accuracy (+0.51 pp), AUC, and min-DCF (2.5× better), yet its EER is marginally worse (6.88% vs 6.56%) with overlapping intervals. The acoustic path trades recall for false alarms (44 FA vs 33 FA for XLS-R). Because min-DCF weights FA 19× more, the acoustic path's 13.8% FA rate is penalised severely.

B. RQ1: Compression Robustness

We compared clean and Opus versions of the same 292 clips (Table IV). The detectors respond oppositely: acoustic drops 4.46 pp (93.84→89.38%); XLS-R rises 1.37 pp (91.44→92.81%). The codec effect is model-dependent.

TABLE IV
RQ1: Clean vs Opus on paired clips (n=292).

Detector	Cond	Acc	EER	F1	mDCF
Acoustic	clean	93.84	5.63	93.53	0.735
Acoustic	Opus	89.38	7.58	89.27	0.765
	Δ	−4.46	+1.95	−4.26	+0.03
XLS-R	clean	91.44	8.13	91.04	0.280
XLS-R	Opus	92.81	6.06	92.19	0.311
	Δ	+1.37	−2.07	+1.15	+0.03

Δ = Opus minus clean.

Score analysis shows bonafide clips shift toward spoof (+0.091 mean score) while spoof clips barely move (−0.013). The codec makes genuine speech look synthetic, not vice versa.

C. RQ2: English→Indic Transfer

The English-only control reaches 54.79% (majority-class) with 76.56% EER, AUC 0.162, min-DCF 1.06. It predicts bonafide for all instances. The main issue is domain shift: features that separate ASVspoof English do not transfer to Indic recordings. On this test set, the model becomes mildly anti-informative.

D. RQ3: Leave-One-Language-Out

Three folds train on two languages and evaluate on the third. Transfer is strongly asymmetric (Table V): Hindi loses 12.16 pp (78.83% vs 90.99% reference), while Marathi and Tamil transfer well (93.29%, 93.94%). Hindi fails through both FA and misses; the other folds fail mainly through FA. Language-independent generalisation is not supported.

TABLE V
RQ3: Leave-one-language-out.

Held out	Train	n	Acc	EER	F1	AUC	Ref
Hindi	mr+ta	222	78.83	21.83 [15.9,26.5]	72.51	0.895	90.99
Marathi	hi+ta	164	93.29	7.14 [2.5,11.3]	95.20	0.989	93.29
Tamil	hi+mr	198	93.94	6.35 [3.3,10.5]	92.21	0.984	90.91

Ref = same language with multilingual detector.

E. RQ4: Calibration

The acoustic path was already well-calibrated (ECE 0.0245); the validation-selected strategy worsened it to 0.0260. XLS-R was overconfident (ECE 0.2048); global temperature reduced it to 0.0891. Calibration benefit is model-dependent. The fine-grained strategy overfit to validation priors that differed from test.

TABLE VI
RQ4: Calibration effect.

Detector	Strategy	val ECE	pre ECE	post ECE	Brier
Acoustic	per-lang-cond	0.0487	0.0245	0.0260	0.069
XLS-R	global T=0.5	0.0686	0.2048	0.0891	0.068

Acoustic: T=0.9/0.5/1.5 for hi/mr/ta.

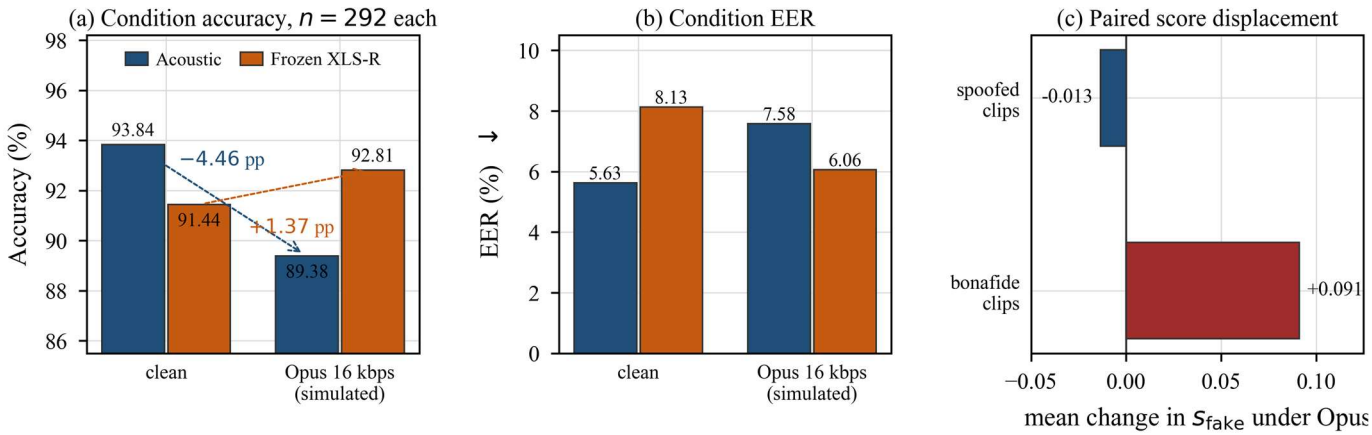


Fig. 3. RQ1 compression. (a) Accuracy and EER by condition;(b) acoustic degrades while XLS-R does not. (c) Score shift: bonafide clips move toward spoof.

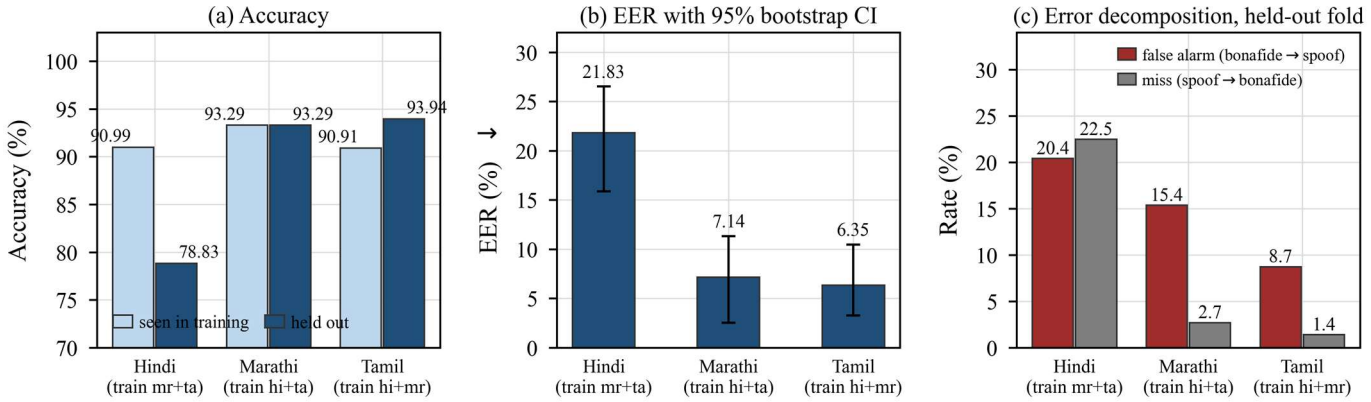


Fig. 4. RQ3 leave-one-language-out. (a) Accuracy vs seen-language reference. (b) EER with 95% bootstrap CI; Hindi is disjoint. (c) Error decomposition.

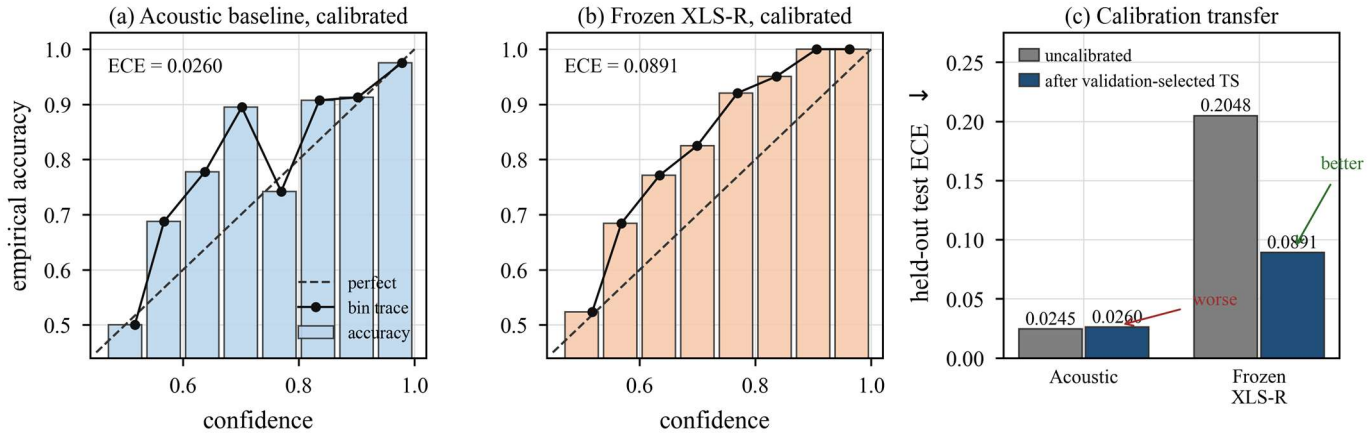


Fig. 5. RQ4 calibration. (a-b) Reliability diagrams. (c) ECE comparison: calibration helped XLS-R but hurt acoustic.

F. Source Confound and V2 Pilot

V1's shortcut probe measured 84.76% label vs 84.59% source—a 0.17 pp gap. But the probe is degenerate: source→label is bijective, so both are the same classifier. The V2 pilot adds 50 FLEURS bonafide clips, breaking bijection. Probes then separate: 85.83% label vs 98.48% source (+12.65 pp). However, FLEURS clips average 10.01 s vs 6.23 s, so duration is partly diagnostic. Table VII summarises status.

TABLE VII
Generalisation status.

Evaluation	n	Status	Outcome
V1 held-out	584	complete	Table III
V2 source probe	593	partial	98.48% vs 85.83%
FLEURS unseen	9	pilot	5/9; CI [21.2, 86.3]%
Generator-disjoint	0	pending	none
Human study	0	blocked	protocol ready

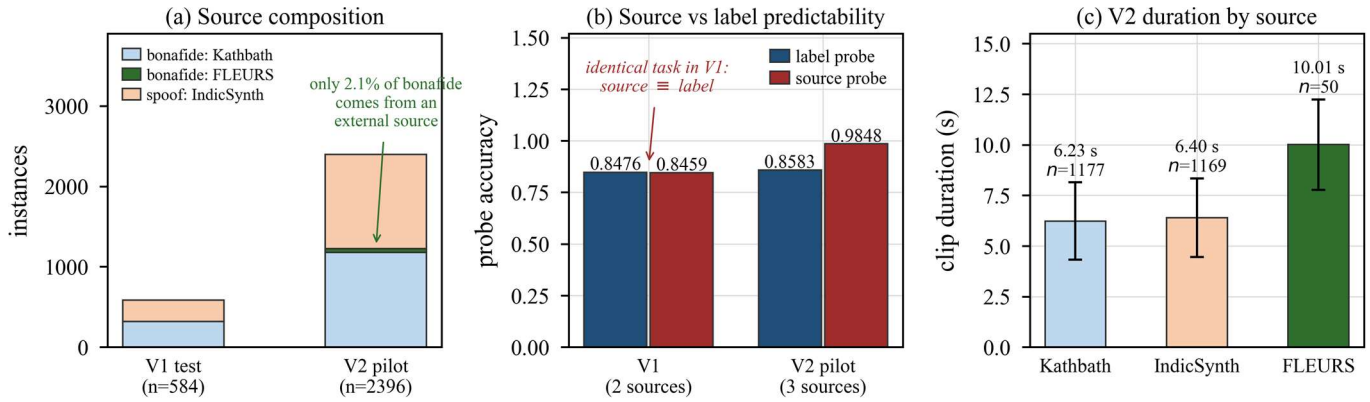


Fig. 6. Source-label confound. (a) V1 vs V2 composition. (b) Source vs label probe: V1 identical; V2 separates by 12.65 pp. (c) FLEURS duration artefact.

VIII. DISCUSSION

Four findings are established: both detectors discriminate well on V1; codec effect is model-dependent; English does not transfer to Indic; cross-lingual transfer is asymmetric. The most important finding is negative: none of these constitutes evidence of source-independent detection.

The 91.61% accuracy measures how well a detector separates held-out speakers from the same corpora. It does not measure generalisation to new sources. A model that learned IndicSynth packaging rather than vocoders would score identically. The V2 source probe (98.48%) and the FLEURS pilot (5/9 correct, 44.4% FA rate vs 13.75% in-domain) point the same way. Until a source-balanced benchmark exists, V1 results should be read as bounded in-domain performance only.

A pattern emerges: the more constrained system transferred better. XLS-R absorbed codec shift while acoustic did not—plausibly because features from 500k hours of heterogeneous audio have seen varied codecs. A global temperature transferred while per-cell temperatures did not—because fine-grained fits overfit validation priors. Both argue for coarser choices when partitions are small.

The operating-point analysis matters for deployment. Accuracy and EER treat error types as interchangeable; deployment does not. A screening system pays for every false alarm in review time and user trust. That is why the acoustic path's 0.7841 min-DCF sits far from XLS-R's 0.3144 despite lower EER. A missed synthetic clip may enable a single fraud; a systematic false-alarm bias against genuine speech from underrepresented accents is a fairness problem that scales with traffic. Our external-source pilot, weak as it is, points at that risk.

A. Reproducibility

Canonical metrics are frozen in artifacts/final_results_manifest.json at commit 084bd47, with per-experiment artefacts, manifests recording source, label, speaker, split, and checksum per clip, and integrity verification via scripts/verify_research_integrity.py. The figures in this paper are regenerated from those artefacts.

IX. LIMITATIONS

Table VIII lists threats to validity. The dominant threat is V1's source-label confound. V2 is incomplete (2.1% external, source probe still 98.48%). Compression is simulated, not WhatsApp. Human comparison is absent (N=0). Generator-disjoint has n=0. Bootstrap intervals span ~4.6 pp for the main cell; smaller differences are not resolved.

TABLE VIII
Threats to validity.

Limitation	Consequence
Source-label confound	May reflect corpus cues
3 of 22 languages	No universal Indic generalisation
Opus simulation	Not WhatsApp transport
V2 incomplete	External-source generalisation unresolved
Generator-disjoint n=0	Unseen-generator robustness unknown
FLEURS n=9	CI [21.2, 86.3]%; uninformative
Human N=0	No human–machine comparison

X. CONCLUSION

VaaniQ evaluates AI-generated voice detection for Hindi, Marathi and Tamil under codec, language, and calibration shifts. Table IX summarises all five RQs. On V1, the acoustic detector reaches 91.61% accuracy, 6.56% EER, 0.9729 AUC and 0.7841 min-DCF; frozen XLS-R reaches 92.12%, 6.88%, 0.9828 and 0.3144. Neither dominates across all metrics. A 16 kbps Opus simulation costs the acoustic path 4.46 pp while leaving XLS-R unharmed. An English-only countermeasure collapses to the majority-class rate on Indic speech. Leave-one-language-out transfer is asymmetric (Hindi 78.83% vs Marathi/Tamil >93%). Validation-selected calibration helped the overconfident XLS-R path but hurt the already-calibrated acoustic path.

TABLE IX
Summary of research questions.

RQ	Experiment	Result	Status
1	Clean vs Opus	Acoustic −4.46 pp; XLS-R +1.37 pp	Done
2	English→Indic	54.79% (majority); AUC 0.162	Done
3	Leave-one-out	Hindi 78.83%; mr/ta >93%	Done
4	Calibration	XLS-R ↓0.0891; Acoustic ↑0.0260	Done
5	Human listeners	N = 0	Blocked

The central message is a constraint on interpretation. V1 draws all bonafide from Kathbath and all spoof from IndicSynth, so corpus identity predicts label perfectly, the source probe is degenerate, and strong in-domain results are compatible with a detector that recognises a dataset rather than a vocoder. The external-source pilot

at 9 clips, the 98.48% V2 source probe, the absence of generator-disjoint evaluation, and the absence of listener data all remain open.

Future work needs: (1) a source-balanced benchmark where bonafide and spoof each come from at least three independent sources; (2) generator-disjoint evaluation; (3) an unseen-source bonafide set of several hundred clips; (4) the listening study in Hindi, Marathi and Tamil. VaaniQ's contribution is not a new detector but a clearer separation between in-domain performance and evidence of generalisation.

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